

Exponential Distribution and the Poisson Process

The Exponential Distribution

A continuous random variable X is said to have an exponential distribution with parameter λ , where $\lambda > 0$, if its probability density function (PDF) is given by:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Equivalently, its cumulative distribution function (CDF) is given by:

$$F(x) = \int_{-\infty}^x f(y)dy = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Mean and Variance

Mean ($E[X]$)

The mean is calculated using integration by parts where $u = x$, $dv = \lambda e^{-\lambda x} dx$:

$$\begin{aligned} E[X] &= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\ &= [-x e^{-\lambda x}]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx \\ &= 0 + \left[-\frac{1}{\lambda} e^{-\lambda x} \right]_0^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

Variance ($Var(X)$)

First, compute $E[X^2]$ using integration by parts ($u = x^2$, $dv = \lambda e^{-\lambda x} dx$):

$$\begin{aligned} E[X^2] &= \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \\ &= [-x^2 e^{-\lambda x}]_0^{\infty} + 2 \int_0^{\infty} x e^{-\lambda x} dx \\ &= 2 \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2} \end{aligned}$$

Now, compute the variance:

$$\begin{aligned} \text{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{1}{\lambda^2} \end{aligned}$$

The Memoryless Property

A random variable X is said to be without memory, or *memoryless*, if:

$$P(X > s + t \mid X > t) = P(X > s) \quad \text{for all } s, t \geq 0$$

This is equivalent to stating:

$$P(X > s + t) = P(X > s)P(X > t)$$

Example:

Suppose the amount of time spent in a bank is exponentially distributed with a mean of 10 minutes ($\lambda = \frac{1}{10}$).

Probability of spending more than 15 minutes:

$$P(X > 15) = e^{-15\lambda} = e^{-3/2} \approx 0.223$$

Probability of spending more than 15 minutes GIVEN she is still there after 10 minutes:
Because of the memoryless property, this is simply the probability of spending at least 5 more minutes:

$$P(X > 5) = e^{-5\lambda} = e^{-1/2} \approx 0.607$$

Properties of Multiple Exponential Variables

Proposition: If $X_1, \dots, X_n$ are independent exponential random variables with a common rate λ , then their sum $\sum_{i=1}^n X_i$ is a gamma (n, λ) random variable with density function:

$$f(t) = \lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}, \quad t > 0$$

Probability of one variable being smaller than another:

If X_1 and X_2 are independent with rates λ_1 and λ_2 :

$$\begin{aligned} P(X_1 < X_2) &= \int_0^\infty P(X_1 < X_2 \mid X_1 = x) \lambda_1 e^{-\lambda_1 x} dx \\ &= \int_0^\infty e^{-\lambda_2 x} \lambda_1 e^{-\lambda_1 x} dx \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \end{aligned}$$

The Poisson Process

Counting Processes:

A stochastic process $\{N(t), t \geq 0\}$ is a counting process if $N(t)$ represents the total number of “events” that occur by time t . It must satisfy:

1. $N(t) \geq 0$
2. $N(t)$ is integer-valued.
3. If $s < t$, then $N(s) \leq N(t)$.
4. For $s < t$, $N(t) - N(s)$ equals the number of events that occur in the interval $(s, t]$.

Independent Increments: A counting process possesses independent increments if the numbers of events that occur in disjoint time intervals are independent.

Little-o Notation (Definition):

The function $f(\cdot)$ is said to be $o(h)$ if:

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$$

Definition (The Poisson Process):

The counting process $\{N(t), t \geq 0\}$ is a Poisson process with rate $\lambda > 0$ if the following axioms hold:

1. $N(0) = 0$
2. The process has independent increments.
3. $P(N(t+h) - N(t) = 1) = \lambda h + o(h)$
4. $P(N(t+h) - N(t) \geq 2) = o(h)$

Interarrival Times and Sequence Distributions

Let T_1 denote the time of the first event, and T_n denote the time between the $(n - 1)$ st and n th event. The sequence $\{T_n, n = 1, 2, \dots\}$ is the sequence of interarrival times.

Lemma:

$T_1, T_2, \dots$ are independent and identically distributed exponential random variables with rate λ .

Proof for T_1 :

$$P(T_1 > t) = P(N(t) = 0) = e^{-\lambda t}$$

Letting $S_n = \sum_{i=1}^n T_i$ be the time of the n th event, it follows from Proposition 5.1 that S_n is a gamma (n, λ) random variable:

$$f_{S_n}(s) = \lambda e^{-\lambda s} \frac{(\lambda s)^{n-1}}{(n-1)!}, \quad s > 0$$

Theorem :

If $\{N(t), t \geq 0\}$ is a Poisson process with rate λ , then $N(t)$ is a Poisson random variable with rate λt . That is:

$$P(N(t) = n) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}, \quad n \geq 0$$